

tables

November 21, 2025

```
[120]: import pandas as pd
        from pathlib import Path

[121]: def load_results_table(results_dir, methods_list, use_coords, use_demographic,
        ↪file_name="aggregated_results.json"):
        agg_list = []
        for method in methods_list:
            if use_demographic and use_coords:
                folder_name = f"{method}_coords_demographic_radiomics"
            elif use_demographic and not use_coords:
                folder_name = f"{method}_demographic_radiomics"
            elif not use_demographic and use_coords:
                folder_name = f"{method}_coords_radiomics"
            else:
                folder_name = f"{method}_radiomics"
            aggregated_result_dir = Path(results_dir)/folder_name / file_name
            df = pd.read_json(aggregated_result_dir)
            agg_list.append(df)
        final_df = pd.concat(agg_list, ignore_index=True)
        final_df = final_df[['model', 'Accuracy_Mean', 'ROC_AUC_Mean']]
        display(final_df)
        print("Average across models:")
        display(pd.DataFrame(final_df.mean(numeric_only=True)).T)
```

1 Five Fold Cross Validation for Center 1 Dataset

```
[122]: print("Radiomics Features:")
        load_results_table("../Results", ["transformer", "mil", "deep_sets", "graph",
        ↪"ml_models"], use_coords=False, use_demographic=False)
```

Radiomics Features:

	model	Accuracy_Mean	ROC_AUC_Mean
0	transformer	0.730931	0.749828
1	mil	0.738027	0.743428
2	deep_sets	0.718257	0.755376
3	graph	0.598183	0.660687

4	AdaBoost	0.724924	0.715395
5	CatBoost	0.710901	0.755721
6	Extra Trees	0.711361	0.762950
7	Gradient Boosting	0.690456	0.710946
8	Logistic Regression	0.676893	0.721450
9	MLP	0.697367	0.763887
10	Random Forest	0.698472	0.780112
11	SVM (RBF kernel)	0.771835	0.809438
12	XGBoost	0.724924	0.759793
13	k-Nearest Neighbors	0.712036	0.731033
14	lightGBM	0.690671	0.733366

Average across models:

	Accuracy_Mean	ROC_AUC_Mean
0	0.706349	0.743561

```
[123]: print("Radiomics + demographic Features:")
load_results_table("../Results", ["transformer", "mil", "deep_sets", "graph",
↪ "ml_models"], use_coords=False, use_demographic=True)
```

Radiomics + demographic Features:

	model	Accuracy_Mean	ROC_AUC_Mean
0	transformer	0.751175	0.866042
1	mil	0.784509	0.860924
2	deep_sets	0.764279	0.837169
3	graph	0.751146	0.834411
4	AdaBoost	0.744494	0.809912
5	CatBoost	0.724709	0.850323
6	Extra Trees	0.718257	0.811522
7	Gradient Boosting	0.737367	0.849030
8	Logistic Regression	0.738057	0.804029
9	MLP	0.750716	0.814383
10	Random Forest	0.725154	0.832488
11	SVM (RBF kernel)	0.778272	0.845303
12	XGBoost	0.718272	0.836490
13	k-Nearest Neighbors	0.725139	0.764889
14	lightGBM	0.690686	0.800226

Average across models:

	Accuracy_Mean	ROC_AUC_Mean
0	0.740149	0.827809

```
[124]: print("Radiomics + coords Features:")
load_results_table("../Results", ["transformer", "mil", "deep_sets", "graph",
↪ "ml_models"], use_coords=True, use_demographic=False)
```

Radiomics + coords Features:

	model	Accuracy_Mean	ROC_AUC_Mean
0	transformer	0.812095	0.882030
1	mil	0.791175	0.835725
2	deep_sets	0.791850	0.860978
3	graph	0.725154	0.755904
4	AdaBoost	0.751175	0.776115
5	CatBoost	0.771175	0.818574
6	Extra Trees	0.725154	0.788914
7	Gradient Boosting	0.723804	0.776966
8	Logistic Regression	0.731146	0.795367
9	MLP	0.763834	0.802898
10	Random Forest	0.725599	0.813257
11	SVM (RBF kernel)	0.792065	0.827623
12	XGBoost	0.777627	0.813542
13	k-Nearest Neighbors	0.704464	0.753744
14	lightGBM	0.764509	0.794150

Average across models:

	Accuracy_Mean	ROC_AUC_Mean
0	0.756722	0.806386

```
[125]: print("Radiomics + coords + demographic Features:")
load_results_table("../Results", ["transformer", "mil", "deep_sets", "graph",
↪ "ml_models"], use_coords=True, use_demographic=True)
```

Radiomics + coords + demographic Features:

	model	Accuracy_Mean	ROC_AUC_Mean
0	transformer	0.818547	0.910149
1	mil	0.811405	0.903351
2	deep_sets	0.825643	0.909966
3	graph	0.804983	0.886705
4	AdaBoost	0.790960	0.857240
5	CatBoost	0.750716	0.869112
6	Extra Trees	0.731821	0.846946
7	Gradient Boosting	0.770701	0.859233
8	Logistic Regression	0.738057	0.831470
9	MLP	0.784968	0.845680
10	Random Forest	0.725154	0.846488
11	SVM (RBF kernel)	0.804509	0.865395
12	XGBoost	0.778057	0.866893
13	k-Nearest Neighbors	0.711361	0.780748
14	lightGBM	0.771405	0.861043

Average across models:

	Accuracy_Mean	ROC_AUC_Mean
0	0.774552	0.862694

2 Second dataset

```
[126]: print("Radiomics Features:")
load_results_table("../Results", ["transformer", "mil", "deep_sets", "graph",
↪ "ml_models"], use_coords=False, use_demographic=False,
↪ file_name="center2_aggregated_results.json")
```

Radiomics Features:

	model	Accuracy_Mean	ROC_AUC_Mean
0	transformer	0.6225	0.655303
1	mil	0.5725	0.602652
2	deep_sets	0.5400	0.580871
3	graph	0.5600	0.579987
4	AdaBoost	0.6025	0.637879
5	CatBoost	0.6325	0.656313
6	Extra Trees	0.6400	0.667740
7	Gradient Boosting	0.5950	0.624242
8	Logistic Regression	0.5875	0.625758
9	MLP	0.6000	0.632828
10	Random Forest	0.6100	0.668308
11	SVM (RBF kernel)	0.4950	0.586742
12	XGBoost	0.6250	0.666162
13	k-Nearest Neighbors	0.5625	0.630682
14	lightGBM	0.6275	0.666667

Average across models:

	Accuracy_Mean	ROC_AUC_Mean
0	0.5915	0.632142

```
[127]: print("Radiomics + demographic Features:")
load_results_table("../Results", ["transformer", "mil", "deep_sets", "graph",
↪ "ml_models"], use_coords=False, use_demographic=True,
↪ file_name="center2_aggregated_results.json")
```

Radiomics + demographic Features:

	model	Accuracy_Mean	ROC_AUC_Mean
0	transformer	0.6625	0.787879
1	mil	0.6300	0.699495
2	deep_sets	0.6225	0.639899
3	graph	0.6375	0.721212
4	AdaBoost	0.6975	0.807576
5	CatBoost	0.7275	0.831187
6	Extra Trees	0.6875	0.764015
7	Gradient Boosting	0.6750	0.792045
8	Logistic Regression	0.6525	0.696970
9	MLP	0.6075	0.692109
10	Random Forest	0.6975	0.786616

11	SVM (RBF kernel)	0.5125	0.602020
12	XGBoost	0.7500	0.852399
13	k-Nearest Neighbors	0.5625	0.637374
14	lightGBM	0.7475	0.832323

Average across models:

	Accuracy_Mean	ROC_AUC_Mean
0	0.658	0.742875

```
[128]: print("Radiomics + coords Features:")
load_results_table("../Results", ["transformer", "mil", "deep_sets", "graph",
↪ "ml_models"], use_coords=True, use_demographic=False,
↪ file_name="center2_aggregated_results.json")
```

Radiomics + coords Features:

	model	Accuracy_Mean	ROC_AUC_Mean
0	transformer	0.6200	0.708712
1	mil	0.6150	0.635101
2	deep_sets	0.5400	0.534154
3	graph	0.6025	0.611869
4	AdaBoost	0.6750	0.739394
5	CatBoost	0.6825	0.718939
6	Extra Trees	0.6425	0.682260
7	Gradient Boosting	0.6850	0.730934
8	Logistic Regression	0.5625	0.627841
9	MLP	0.6125	0.673422
10	Random Forest	0.6425	0.684091
11	SVM (RBF kernel)	0.5000	0.578283
12	XGBoost	0.6675	0.752399
13	k-Nearest Neighbors	0.5650	0.632513
14	lightGBM	0.6925	0.764773

Average across models:

	Accuracy_Mean	ROC_AUC_Mean
0	0.620333	0.671646

```
[129]: print("Radiomics + demographic + coords Features:")
load_results_table("../Results", ["transformer", "mil", "deep_sets", "graph",
↪ "ml_models"], use_coords=True, use_demographic=True,
↪ file_name="center2_aggregated_results.json")
```

Radiomics + demographic + coords Features:

	model	Accuracy_Mean	ROC_AUC_Mean
0	transformer	0.7075	0.811364
1	mil	0.6700	0.741793
2	deep_sets	0.6775	0.753030
3	graph	0.7300	0.781187

4	AdaBoost	0.7675	0.833460
5	CatBoost	0.7275	0.825000
6	Extra Trees	0.7025	0.785038
7	Gradient Boosting	0.7375	0.812247
8	Logistic Regression	0.6500	0.694697
9	MLP	0.6400	0.711174
10	Random Forest	0.6800	0.787311
11	SVM (RBF kernel)	0.5150	0.601957
12	XGBoost	0.7875	0.858838
13	k-Nearest Neighbors	0.5650	0.639520
14	lightGBM	0.7800	0.854419

Average across models:

	Accuracy_Mean	ROC_AUC_Mean
0	0.689167	0.766069

3 Test Time Adaptation (TTA) on Center 2 data

3.1 TENT method for TTA

```
[130]: print("Radiomics + demographic + coords Features:")
load_results_table("../Results", ["transformer", "mil", "deep_sets", "graph"],
    ↪use_coords=True, use_demographic=True,
    ↪file_name="tta_tent_center2_aggregated_results.json")
```

Radiomics + demographic + coords Features:

	model	Accuracy_Mean	ROC_AUC_Mean
0	transformer	0.7100	0.868308
1	mil	0.7700	0.864141
2	deep_sets	0.7725	0.857702
3	graph	0.6900	0.786111

Average across models:

	Accuracy_Mean	ROC_AUC_Mean
0	0.735625	0.844066

3.2 Norm method for TTA

```
[131]: print("Radiomics + demographic + coords Features:")
load_results_table("../Results", ["transformer", "mil", "deep_sets", "graph"],
    ↪use_coords=True, use_demographic=True,
    ↪file_name="tta_norm_center2_aggregated_results.json")
```

Radiomics + demographic + coords Features:

	model	Accuracy_Mean	ROC_AUC_Mean
0	transformer	0.7075	0.811364
1	mil	0.6700	0.741793

2	deep_sets	0.6775	0.753030
3	graph	0.7300	0.781187

Average across models:

	Accuracy_Mean	ROC_AUC_Mean
0	0.69625	0.771843

[]: